One Stream, Many Windows: Soft Targets for Consolidating Fast and Slow Conversions

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Abstract. Conversion pipelines in advertising are split by speed. An upper-funnel onsite conversion is observable within minutes of the click, while a down-funnel offsite conversion takes a day or more. Each objective defines an *attribution window*, the elapsed time within which a conversion counts, and conventionally that window also fixes the *join delay*, the age at which the click's training record is written. Tying the two together forces every objective onto its own pipeline, each as stale as the window it waited for. We ask whether they can be set independently: can an objective accountable to a seven-day window be trained from records joined within 90 minutes?

Writing records sooner labels every conversion still in flight as a negative, and a streaming pipeline that trains each click at most once cannot revise them afterwards, yet most published corrections do exactly that: of 11 methods we reproduce, 8 write more than one record per click, up to 4.00. The deficit must instead be filled when the record is written, by an estimate from a model trained on matured outcomes.

On the Criteo Delayed Feedback Dataset we compare three targets across 6,900 runs: the matured label, denoted Wait; the teacher's estimate alone, Distill-Only; and the observed label with only its negatives softened, Correct-Distill. We vary served capacity over 1,890 to 9,580,477 parameters, three join delays, two attribution windows, and two teacher architectures differing 6-fold in size. Against a Wait baseline of NE 0.72 to 0.81, Distill-Only reduces NE by 0.60% at the two smallest capacities but increases it by 0.51% at the three largest. Correct-Distill improves on Distill-Only in all 30 configurations and on Wait in 28, reducing NE by 1.97% at seven days and 0.57% at one, which is 74% and 69% of what a label-peeking oracle attains. It is also the strongest method that obeys the one-record rule, and the larger teacher does not help: it changes the result by 0.043 pp. Setting the join delay independently of the attribution window therefore removes the obstacle to training objectives with different windows on a single fresh stream.

Keywords: Conversion rate prediction · Delayed feedback · Online advertising · Knowledge distillation · Streaming training pipelines.

1 Introduction

Advertising systems predict several kinds of conversion at once, and those kinds differ sharply in how long they take to observe. A view or an add-to-cart is

recorded within minutes of the click; a purchase, an app install or a submitted lead may take a day or a week. Each objective is defined with an *attribution window*, the elapsed time within which a conversion is credited to the click, and a production system maintains several windows at once.

Conventionally that window also determines the *join delay*, the age at which the click's training record is written to the stream: the pipeline waits until the label is knowable, then emits the record. The convention costs twice. A model for a slow objective trains only on data at least a window old, so it never sees recent behaviour. And a system that ranks several objectives jointly must either maintain one pipeline per objective or delay all of them until the slowest window closes, imposing the staleness of the slowest funnel on the fastest.

Neither cost is intrinsic. The attribution window is fixed by the business definition of a conversion; the join delay is a pipeline choice. We ask what happens when the two are set independently, and specifically whether an objective accountable to a seven-day window can be trained from records joined within 90 minutes.

The obstacle is that an early record is wrong in a known direction: every conversion still in flight when the record is written is recorded as a negative. A substantial literature corrects this censoring bias [1,11,10,2,3], but most corrections revise the record afterwards, by duplicating the click when its outcome lands, by relabelling it, or by adding prediction heads at further horizons. A streaming pipeline that appends each record once and trains on it once can do none of these. We state that restriction as a *single-visit* constraint and measure which published methods satisfy it. What then remains is a choice of what the single frozen target should contain, and the deficit must be filled at write time by an estimate from a model that has seen matured outcomes. Such a model is readily available in production, where large ranking models are already trained across objectives on fully matured data [7].

We make three contributions. **First**, we separate the join delay from the attribution window, state the causality and single-visit rules a deployable pipeline must satisfy, and audit 11 published methods against them; 8 violate single-visit, writing up to 4.00 records per click. **Second**, we show that writing the record early and softening only its observed negatives beats waiting for the matured label, by more as the funnel slows, and is the strongest method obeying the single-visit rule. **Third**, we show the result needs neither a large teacher, an expensive one, nor one retrained while the student trains, which is what makes it cheap to adopt beside an existing ranking system.

2 Setting and the Single-Visit Constraint

A click i occurs at time C_i with features x_i and, if it converts, does so after a delay $D_i > 0$, with $D_i = \infty$ when it never converts. The attribution window v belongs to the objective and fixes the target $Y_i^{(v)} = \mathbf{1}[D_i \le v]$, knowable only at $C_i + v$. The join delay o belongs to the pipeline and fixes when the record is

written, so the label available at that moment is

$$Y_i^{(o)} \;=\; \mathbf{1}[D_i \le o] \;\le\; Y_i^{(v)}. \tag{1}$$

Setting $o = v$ reproduces the conventional per-objective pipeline. Setting $o \ll v$ is what we want, and the inequality in Eq. (1) is its price. We write $F(o) = P(D \le o \mid D \le v)$ for the fraction of the objective's conversions already visible when the record is written.

Training proceeds in hourly rounds. Each round consumes the records released since the previous cut-off, then scores every click of the following hour exactly once against the matured target $Y^{(v)}$. Every method is therefore judged on the business label whatever it trained on, and updating before predicting keeps the evaluation causal.

Two rules. A deployable pipeline must satisfy two rules.

Rule 1, causality. A record's label must be computable at the moment the record is written, and must not change afterwards. Assigning $Y_i^{(v)}$ to a record written at $C_i + o$ uses an outcome that has not yet occurred. Such an arm is an upper reference, not a system, and we report it under the name Oracle.

Rule 2, single-visit. Let n_i be the number of times click i enters the served model's loss over its lifetime. Rule 2 requires $\max_i n_i \le 1$. Without it the question is empty, since any join delay can be shortened by writing a record early and writing it again later.

We take the maximum over clicks rather than the mean, because it is the existence of a second write that obliges the pipeline to support revision, and the rule governs the served stream only: charging training-time auxiliaries would eliminate Fsiw, Ulc and our own arms alike.

Which published methods comply. We reproduced 11 published methods from scratch under a single protocol and counted the records each one's served model consumes per distinct click. 8 write more than one.[1] At $v = 1\,\mathrm{d}$ the counts are FTP 4.00, MISS 3.05, IF-DFM 2.79, PI 2.00, DDFM 1.89, DFM 1.33, DEFUSE 1.03, ES-DFM 1.03 . The highest, IF-DFM, reaches 97 writes for a single click, its mechanism being an update applied whenever a label reverses. The plain re-ingestion baseline [5,4] measures 1.03 and fails the same criterion. Duplication is a property of emission logic rather than of the model, and the audit returns identical counts at the smallest and largest served capacities.

Only FSIW, TWICE, ULC satisfy Rule 2. Of these, Fsiw corrects the loss and Twice adds a prediction head on the conversion clock, both leaving the target itself untouched. Ulc alone changes the target, which makes it our closest comparator. Section 5 reports how all of them perform, rather than only whether they comply.

[1] Two measurement traps both understate duplication. The denominator must be distinct clicks touched, since dividing records consumed by records released conflates duplication with variation in click volume. And a scoring window shorter than a method's revisit horizon cannot observe the revisit, so we score only clicks whose entire horizon lies inside the stream.

3 Four Targets

Writing the record early leaves a deficit: conversions arriving between o and v are recorded as negatives, and something must stand in for them in the record's single frozen target. We study four answers. All four share a pipeline, schedule, feature set and optimiser, and differ only in the target $\tilde{y}_i$ and in the age Δ at which the record is written.

FRESH writes at $\Delta = o$ with $\tilde{y}_i = Y_i^{(o)}$, filling the deficit with nothing. WAIT writes at $\Delta = v$ with $\tilde{y}_i = Y_i^{(v)}$, avoiding the deficit by deferring the write. It has a correct label on data of age v, and it is the conventional per-objective pipeline, so it is our baseline throughout. A practitioner who refuses both to duplicate records and to train on wrong labels is already running it.

CORRECT-DISTILL writes at $\Delta = o$, keeps the observed label and softens only the observed negatives:

$$\tilde{y}_i \;=\; Y_i^{(o)} + (1 - Y_i^{(o)})\, w^T(x_i), \qquad w^T(x) \approx P\Big(Y^{(v)}{=}1 \,\Big|\, Y^{(o)}{=}0,\, x\Big)\,. \tag{2}$$

An observed positive stays exactly 1, and only the uncertain mass moves. This is the functional form used by ULC [8]; the two differ in how w^T is estimated, so comparing them isolates the estimator and nothing else.

DISTILL-ONLY goes further and regresses on $\hat{p}_v(x) \approx P(Y^{(v)}{=}1 \mid x)$ alone, so the record carries features and a number but no observation at all.

The teacher. The estimates w^T and $\hat{p}_v$ come from a separate model trained on the *matured* stream, in which a record enters at $C_i + v$ carrying $Y_i^{(v)}$. Both quantities are read out from one shared trunk. The teacher is *evaluated* at the instant the student's record is written: at $C_i + o$ the student queries it and freezes the returned number into the record. Rule 1 is therefore satisfied even though the target is not binary. Softness is a different way of satisfying causality, not an exemption from it. The construction is distillation from privileged information [9], in which the teacher observes the matured label that the student by construction cannot.

Being used only during training, the teacher costs nothing at serving time, though it does mean served capacity is matched across arms while total training capacity is not. We therefore treat teacher architecture as an experimental axis in Section 4.

We also distinguish two ways of maintaining the teacher. A *continuously retrained* teacher keeps consuming the matured stream throughout the run, so deploying it means operating a second training job alongside the first. A *fixed* teacher is trained once during causal initialisation and never updated, so deploying it means shipping a static model file. The distinction is invisible in Eq. (2) and decisive operationally, so we measure both.

Finally, ORACLE writes at $\Delta = o$ but labels the record $Y_i^{(v)}$. It violates Rule 1 and is not deployable. We include it because it bounds what any causal correction could recover, which is what turns a positive result into a statement about remaining headroom.

4 Experimental Setup

Data and protocol. We use the public Criteo conversion logs released with [1], the only large public advertising corpus carrying per-click conversion timestamps. Clicks span 60 days. Days $[0, 30)$ initialise the model causally and days $[30, 60)$ form the streaming period, divided into 719 hourly rounds yielding 8,196,131 scored clicks at a base conversion rate of 0.1361. Learning rates are selected per arm and per cell from a seven-point grid on a held-out slice of the initialisation period, before any streaming evaluation, then pinned across seeds. The teacher's rate is fixed and decoupled from the student's, so the student's sweep tunes one model rather than two.

Axes. Two attribution windows, $v = 1\,\mathrm{d}$ and $v = 7\,\mathrm{d}$, stand for a fast and a slow funnel; the contrast between them is the paper's point, so we never average across them. Three join delays are used, $o \in \{5, 30, 90\}$ minutes. Five served capacities range from 1,890 to 9,580,477 parameters and 0.001 to 0.286 MFLOPs per inference. Parameters and compute do not track each other, since the two smallest rungs differ 5.2-fold in parameters at identical compute, so we report both rather than let one stand for the other.

Two teacher architectures. The fourth axis is the teacher, and we study two configurations differing in architecture rather than only in scale. The *small* teacher places 8-dimensional embedding tables under a two-layer perceptron, giving 9,456,897 parameters at 0.039 MFLOPs. The *large* teacher places 48-dimensional tables under a deep interaction trunk, giving 57,272,515 parameters at 1.593 MFLOPs, a 6-fold increase in size and 41-fold in compute. Reporting both is what licenses any claim about whether teacher capacity matters.

Scale and comparability. The evidence comprises 6,900 runs in two campaigns: 4,850 runs over 37 arms reproducing the published methods and establishing the small-teacher results, and 2,050 runs over 14 arms studying the three targets at the large teacher, with 10 seeds on the headline arms and 5 elsewhere. The second campaign runs a frozen copy of the first's harness. WAIT instantiates no teacher, so nothing the second campaign changed can reach it, and it reproduces the first to zero difference across all 150 shared cells. Reporting every arm as a gain against WAIT within its own campaign is therefore what places both on a single axis, rather than an assumption that two runs are comparable.

Metric and statistics. We report normalised entropy against $Y^{(v)}$, lower being better, pooled over all scored clicks rather than averaged over rounds, since a per-round average would normalise each hour by its own base rate and discard diurnal variation a model can legitimately predict. Results are the relative reduction against WAIT within a cell, computed per seed and then averaged. Seeds differ only in initialisation and minibatch order, so seed i of two arms forms a matched pair: we use a paired t-test within each cell and Holm–Bonferroni correction across the 15 cells of a contrast, counting only cells whose gain is

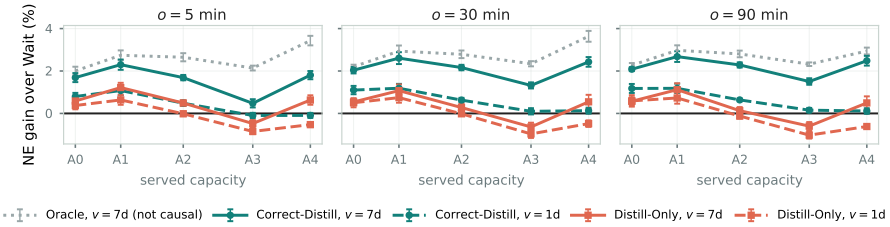


Fig. 1. NE gain over Wait against served capacity, one panel per join delay. Above the line beats the conventional per-objective pipeline. Colour denotes the target and dash denotes the attribution window, with Oracle drawn at $v = 7\,\mathrm{d}$ to bound the panel. Bars give the standard deviation of the per-seed differences.

positive and survives. Seed dispersion is small away from Fresh, with median NE standard deviation 0.00107 and maximum 0.00520. Fresh reaches 0.0490 at $o = 5$ minutes, a finding rather than a nuisance: an uncorrected censored target at the shortest join delay is not a stable estimator. Seeds measure initialisation variance on one data origin, establishing precision rather than generalisation.

5 Results

Throughout, the reported quantity is the relative NE reduction against Wait, so a positive number means the short join delay is better than waiting for the matured label.

A short join delay beats waiting, and by more on slow funnels. Writing the record 2,016 times sooner than the seven-day window it answers to, and letting the teacher stand in for the conversions not yet seen, reduces NE in all 15 seven-day cells, of which 15 survive Holm–Bonferroni correction. The reduction ranges from 0.48% to 2.68%, with a mean of 1.97%. At the one-day window the same substitution gains 0.57% on average, positive in 13 of 15 cells and significant in 10.

The two windows differ by a factor of three, and the direction is the informative part. The slower the funnel, the more a short join delay is worth. This inverts the natural expectation, since a longer window censors more of the label and leaves the teacher more to supply. What dominates instead is the cost of the alternative: Wait at $v = 7\,\mathrm{d}$ trains on week-old data, and no amount of label accuracy compensates for that.

Both ends of the achievable interval are measured. Oracle, which writes at o but labels the record $Y^{(v)}$, gains 2.68% and 0.83%, so Correct-Distill recovers 74% and 69% of what any causal correction could attain. Fresh, the same early write with no correction at all, loses 20.1% and 11.8%. The deficit the teacher fills is an order of magnitude larger than the gains above it.

Table 1. Mean NE gain over WAIT in percent, across the 15 cells of each window.

Method	Single-visit	v=1 d	v=7 d
DEFUSE	no	+0.75	+2.23
FSIW [11]	yes	-0.45	+0.31
TWICE	yes	-28.89	-29.71
ULC [8]	yes	+0.56	+1.84
CORRECT-DISTILL (ours)	**yes**	**+0.57**	**+1.97**

The record must keep part of its hard label. The result above uses the hybrid target of Eq. (2). The fully soft target decides how far it can be pushed. At $v = 1$ d, DISTILL-ONLY beats WAIT by 0.60% at the two smallest capacities and loses to it by 0.51% at the three largest, crossing over between A1 and A2 at every join delay as Fig. 1 shows. A purely soft target plausibly acts as a regulariser, supplying a variance reduction that a small model can use and a larger one no longer needs [6]. The practical consequence cuts against convenience: a record carrying no hard label is the most portable object a consolidated pipeline could ask for, and it is exactly the one that fails at the capacities worth serving. At $v = 7$ d the crossover is weaker and non-monotone, averaging 0.10% across the three largest rungs, so we state the rule for the one-day window and do not extrapolate it. CORRECT-DISTILL beats DISTILL-ONLY in all 30 cells, and WAIT in 28 of 30. The 2 exceptions both occur at $v = 1$ d and $o = 5$ minutes, at rungs A3, A4, the worse by 0.097%, and neither survives correction. Their location is coherent rather than random: the shortest join delay at the fast window is where the teacher has least to add, since 0.355 of the conversions are already visible.

Comparison with published methods. Table 1 reports how the screened methods perform, rather than only whether they comply.

Two conclusions follow. First, the one-record rule is inexpensive: only DEFUSE outperforms our method, and it writes 1.03 records per click to do so, buying 0.26 pp at seven days and 0.18 pp at one. Second, among methods that can actually run on such a stream, ours is the strongest. ULC [8] shares Eq. (2) and differs only in how the corrector is estimated; we lead it by +0.13 pp at seven days, winning 12 of 15 cells, and match it at one day, winning 10 of 15. The case for an offline teacher over an online corrector is therefore architectural rather than one of accuracy, as Section 6 argues. TWICE does not reproduce competitively under our protocol. We report it for completeness and make no claim about its published standing; its mechanism behaves as described, but the floor it is read against sits well above every other method's in our harness.

The teacher need not be large, costly or retrained. Table 1 uses the large teacher. Substituting the small one changes little. CORRECT-DISTILL gains 2.00% with the small teacher against 1.97% with the large one at seven days, and 0.63% against 0.57% at one. Averaged over all 30 cells the large teacher moves the result by -0.043 pp for CORRECT-DISTILL and +0.114 pp for DISTILL-ONLY. A

6-fold increase in teacher parameters and a 41-fold increase in teacher compute therefore buy nothing measurable, and the smaller teacher is if anything marginally ahead.

Holding parameters fixed while spending 4.7 times more teacher compute is equally inert. The teacher's learning rate is verified rather than inherited: the default beats a rate ten times smaller in every cell tested and matches one twice as large, so the null result reflects a saturated teacher rather than an untuned one.

Finally, fixing the teacher after causal initialisation, so that it is shipped as a static model rather than maintained as a second training job, costs 0.08 and 0.27 pp. The fixed-teacher arm still beats WAIT in 27 of 30 cells. Adoption therefore does not require standing up a continuously retrained auxiliary alongside the served model.

6 Implications for Consolidated Pipelines

Training several ranking objectives in one model is an active industrial direction [7], and the obstacle it meets first is a pipeline obstacle rather than a modelling one. Under WAIT, objective k's record exists only at $C_i + v_k$. A shared append-only stream must therefore either write K records per click, violating Rule 2 K times over, or delay every objective until $\max_k v_k$, imposing the slowest funnel's staleness on the fastest.

Our result removes that dilemma. A single stream at one short join delay can carry

$$\tilde{y}_{i,k} \;=\; Y_{i,k}^{(o)} + (1 - Y_{i,k}^{(o)})\, w_k^T(x_i) \tag{3}$$

for every objective at once, because w_k^T is a function the pipeline evaluates rather than an outcome it awaits. Doing so improves accuracy rather than trading it away, and improves it most for the slow objectives that forced the separation in the first place.

ULC is the closest alternative and reaches comparable accuracy, so the choice between them is architectural. ULC fits each w_k inside the served model's training loop, turning K objectives into K auxiliary heads on one optimiser. A teacher instead places each w_k upstream, with a single number written onto a record as its only interface. Section 5 establishes what that upstream model has to be, and the answer is undemanding on every count that matters for adoption: not large, not computationally expensive, and not retrained while the student trains. Consolidation therefore adds K static model files to the build rather than K running training jobs. This is the regime in which an existing cross-objective ranking model is attractive as the source of w_k^T, not because it is large but because it already exists, already observes matured outcomes for every objective, and can be queried once per record and then ignored.

What this work does not establish. We ran one objective at a time, so we have no evidence on interference between objectives that share a trunk. That is the

central risk in Eq. (3) and the thing a multi-task study must measure. Our teacher also stands in for the production ranking model the argument invokes rather than being an instance of one. Finally, one corpus with one chronological split measures precision rather than generalisation.

Conclusion. An objective's attribution window is fixed by the business definition of a conversion, but the join delay its pipeline adopts is not. The two can be set independently. Shortening the join delay by three orders of magnitude improves accuracy rather than degrading it, recovers 74% of what a label-peeking oracle attains at seven days, and helps most where the funnel is slowest, provided the record retains part of its hard label. The fully label-free target, which would be the most convenient object for a consolidated pipeline to carry, is the one that fails. The experiment this makes worth running is K prediction heads, K attribution windows, one stream, and the teachers placed upstream.

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